

Assignment 2 - Basics of Control Theory

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Evaluation is mostly based on how clearly you explain concepts and motivate statements, rather than computations. If you show the correct procedure but you make a mistake in the computation, it won't affect how the exercise is evaluated, as long as the procedure is properly motivated.

Exercise 1 (50 Points)

Consider the dynamical matrix system of exercise 3 of the previous homework:

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix},$$

with

$$B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 0 & 1 \end{bmatrix}.$$

1. Does a control law $u = -Kx$, with $K = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \end{bmatrix}$ stabilize the system?
(25 points)

2. Now consider instead $B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$. Find the minimal dimension of the controller $G(s)$ that allows eigenvalue assignment. (25 points)

Exercise 2 (50 Points)

Fill in the python script `hw02.py`. You're required to simulate the evolution of the system in exercise 1.1 and visualize the evolution of the state variables when using $k = \{1, 3, 5\}$. What do we expect in each case?